

Master Theorem Worksheet

This is a worksheet to help you master solving recurrence relations using the Master Theorem. For each recurrence, either give the asymptotic solution using the Master Theorem (state which case), or else state that the Master Theorem doesn't apply. You should be able to go through these 25 recurrences in 10 minutes.

Problem 1-1. $T(n) = 3T(n/2) + n^2$

$a = 3$

$b = 2$

$f(n) = n^2$

Problem 1-2. $T(n) = 7T(n/2) + n^2$

$a = 7$

$b = 2$

$f(n) = n^2$

Problem 1-3. $T(n) = 4T(n/2) + n^2$

$a = 4$

$b = 2$

$f(n) = n^2$

Problem 1-4. $T(n) = 3T(n/4) + n \lg n$

$a = 3$

$b = 4$

$f(n) = n \lg n$

Problem 1-5. $T(n) = 4T(n/2) + \lg n$

$a = 4$

$b = 2$

$f(n) = \lg n$

Problem 1-6. $T(n) = T(n-1) + n$

can't

doesn't apply?

$\Theta(n^2)$

$\log_2 3 > 1$

$n^2 = \Theta(n^{\log_2 3})$

$\Theta(n^2)$

$\log_2 7 > 1$

$n^2 = \Theta(n^{\log_2 7})$

$\Theta(n^{\log_2 7})$

$n^2 = \Theta(n^{\log_2 4})$

$\Theta(n^2)$

$\log_2 4 > 1$

$n^2 = \Theta(n^2)$

$\Theta(n^2)$

$\Theta(n^2 \lg n)$

$\log_4 3 < 1$

$n \lg n = \Omega(n^{\log_4 3})$

$\Theta(n \lg n)$

$\log_2 4 = 2$

$\lg n = O(n^2)$

$\Theta(n^2)$

Problem 1-7. $T(n) = 4T(n/2) + n^2 \lg n$

$$a = 4$$

$$b = 2$$

$$\log_2 4 = 2$$

$$n^2 / \log_2(n) = \Omega(n^2)$$

$$\boxed{\Theta(n^2 \log_2(n)^2)}$$

$$f(n) = n^2 / \log_2(n)$$

Problem 1-8. $T(n) = 5T(n/2) + n^2 \lg n$

$$a = 5$$

$$b = 2$$

$$2 < \log_2 5 < 3$$

$$n^2 \lg(n) = O(n^{\log_2 5})$$

$$\boxed{\Theta(n^{\log_2 5})}$$

$$f(n) = n^2 \lg(n)$$

Problem 1-9. $T(n) = 3T(n/3) + n / \lg n$

$$a = 3$$

$$b = 3$$

$$\log_3 3 = 1$$

$$f(n) = \frac{n}{\lg n}$$

$$\cancel{\Theta(n)} \quad \cancel{\Theta(n)} \quad \boxed{\text{doesn't apply}}$$

Problem 1-10. $T(n) = 2T(n/4) + c$

$$a = 2$$

$$b = 4$$

$$\log_4 2 = \frac{1}{2}$$

$$c = O(\sqrt{n})$$

$$\boxed{\Theta(\sqrt{n})}$$

$$f(n) = c$$

Problem 1-11. $T(n) = T(n/4) + \lg n$

$$a = 1$$

$$b = 4$$

$$\log_4 1 = 0$$

$$\lg(n) = \Omega(1)$$

$$\boxed{\Theta(\lg^2 n)}$$

$$f(n) = \lg n$$

Problem 1-12. $T(n) = T(n/2) + T(n/4) + n^2$

$$a = 2$$

$$\boxed{\text{doesn't apply}}$$

Problem 1-13. $T(n) = 2T(n/4) + \lg n$

$$a = 2$$

$$b = 4$$

$$\log_4 2 = \frac{1}{2}$$

$$\lg(n) = O(\sqrt{n})$$

$$\boxed{\Theta(\sqrt{n})}$$

$$f(n) = \lg n$$

Problem 1-14. $T(n) = 3T(n/3) + n \lg n$

$$a = 3$$

$$b = 3$$

$$\log_3 3 = 1$$

$$n \lg n = \Omega(n)$$

$$\boxed{\Theta(n \lg^2(n))}$$

$$f(n) = n \lg n$$

Problem 1-15. $T(n) = 8T(n/4) + n^2$

602

$\boxed{\text{Master applicable}}$

Problem 1-16. $T(n) = 2T(n/4) + \sqrt{n}$

$a = 2$
 $b = 4$

$\log_4 2 = \frac{1}{2}$

$\sqrt{n} = \Theta(\sqrt{n})$

$\boxed{\Theta(\sqrt{n} \log n)}$

$f(n) = \sqrt{n}$

Problem 1-17. $T(n) = 2T(n/4) + n^{0.51}$

$a = 2$

$b = 4$

$\log_4 2 = \frac{1}{2}$

$n^{0.51} = \Omega(n^{0.5})$

$\boxed{\Theta(n^{0.51})}$

$f(n) = n^{0.51}$

Problem 1-18. $T(n) = 16T(n/4) + n!$

$a = 16$

$b = 4$

$\log_4 16 = 2$

$n! = \Omega(n^2)$

$\boxed{\Theta(n!)}$

$f(n) = n!$

Problem 1-19. $T(n) = 3T(n/2) + n$

$a = 3$

$b = 2$

$\log_2 3 < 2$

$n = O(n^{\log_2 3})$

$\boxed{\Theta(n^{\log_2 3})}$

$f(n) = n$

Problem 1-20. $T(n) = 4T(n/2) + cn$

$a = 4$

$b = 2$

$\log_2 4 = 2$

$cn = O(n^2)$

$\boxed{\Theta(n^2)}$

$f(n) = cn$

Problem 1-21. $T(n) = 3T(n/3) + n/2$

$a = 3$

$b = 3$

$\log_3 3 = 1$

$\frac{n}{2} = \Theta(n)$

$\boxed{\Theta(n \log n)}$

$f(n) = \frac{n}{2}$

Problem 1-22. $T(n) = 4T(n/2) + n/\lg n$

$a = 4$

$b = 2$

$\log_2 4 = 2$

$\frac{n}{\lg n} = O(n^2)$

$\boxed{\Theta(n^2)}$

$f(n) = \frac{n}{\lg n}$

Problem 1-23. $T(n) = 7T(n/3) + n^2$

$$a = 7$$

$$b = 3$$

$$1 < \log_3 7 < 2$$

$$n^2 = \Omega(n^{1.7})$$

$$\boxed{\Theta(n^2)}$$

Problem 1-24. $T(n) = 8T(n/3) + 2^n$

$$a = 8$$

$$b = 3$$

$$1 < \log_3 8 < 2$$

$$2^n = \Omega(n^{10.5})$$

$$\boxed{\Theta(2^n)}$$

Problem 1-25. $T(n) = 16T(n/4) + n$

$$a = 16$$

$$b = 4$$

$$\log_4 16 = 2$$

$$n = O(n^2)$$

$$\boxed{\Theta(n^2)}$$

$$f(n) = n$$

~~Master Theorem~~

Merge: $T(n) = 2T(\frac{n}{2}) + n$ $\log_2 2 = 1$ $n = \Theta(n), \Theta(n \log n)$

~~Master Theorem~~